DEP521E – HW2 Part A
Bilinear Element Model and Testing

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1 Method

The bilinear element has a primary stiffness k_p and a secondary stiffness $k_s = k_2$:

$$k_p = \frac{f_y}{d_y}, \qquad q_y = f_y - k_2 d_y$$

The Top and Bottom Borders are the lines with slope k_2 through (d_y, f_y) and $(-d_y, -f_y)$:

$$F_{top}(x) = q_y + k_2 x, \qquad F_{bot}(x) = -q_y + k_2 x$$

The force always stays between the two borders. The state parameter shows where the element is: 0 = Interior Region, 1 = Top Border, -1 = Bottom Border. For a step Δx from (x_i, f_i) the borders are evaluated at $x_{i+1} = x_i + \Delta x$ and:

- **Interior Region:** $\Delta f = k_p \Delta x$. If $f_i + \Delta f$ goes beyond the Top (Bottom) Border, then $\Delta f = F_{top} - f_i$ ($F_{bot} - f_i$) and the state becomes Top (Bottom). Otherwise the state stays Interior.
- **Top Border:** if $\Delta x \geq 0$, $\Delta f = k_s \Delta x$ and the state stays Top. If $\Delta x < 0$, the element unloads with $\Delta f = k_p \Delta x$ and the state becomes Interior (or Bottom, if it goes beyond the Bottom Border).
- **Bottom Border:** same rules in the opposite direction.

The total force is $f_{i+1} = f_i + \Delta f$.

2 Code

- `bilinear(dy,fy,k2,x_ini,f_ini,dx,param_a)`: applies the rules above and returns the incremental force `df` and the new state `param_a`.
- Main part: starts from $x = 0$, $f = 0$, `param_a = 0`, calls `bilinear` for each step of the displacement history, adds `df` to the force and plots the results.
- Parameters: $d_y = 1$, $f_y = 5$, $k_2 = 0.25$, so $k_p = 5$ and $q_y = 4.75$.
- Input displacement: the 24-point history given in the assignment ($\Delta t = 1$ is used for plotting).

3 Results

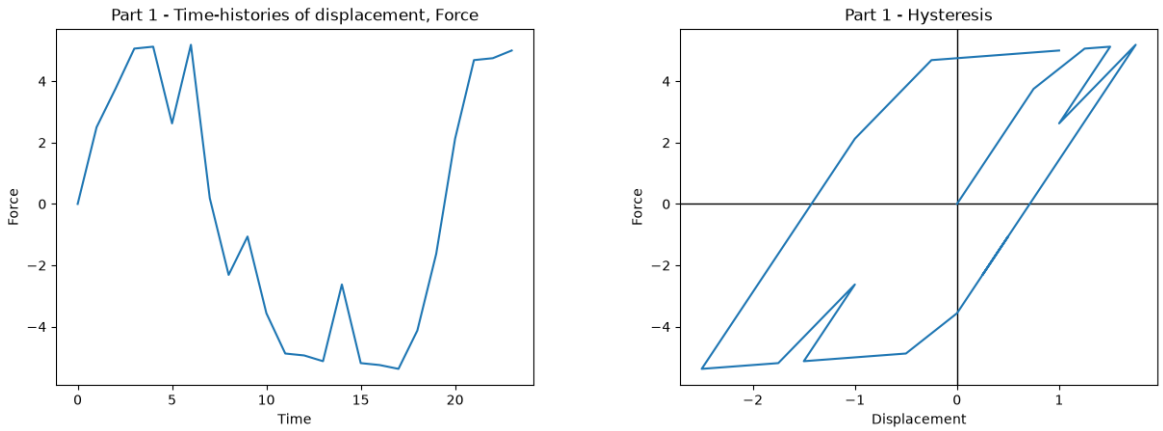


Figure 1: Force time history (left) and hysteresis: displacement vs force (right).

Step	x	f	State	Step	x	f	State
0	0.00	0.0000	Interior	12	−0.75	−4.9375	Bottom
1	0.50	2.5000	Interior	13	−1.50	−5.1250	Bottom
2	0.75	3.7500	Interior	14	−1.00	−2.6250	Interior
3	1.25	5.0625	Top	15	−1.75	−5.1875	Bottom
4	1.50	5.1250	Top	16	−2.00	−5.2500	Bottom
5	1.00	2.6250	Interior	17	−2.50	−5.3750	Bottom
6	1.75	5.1875	Top	18	−2.25	−4.1250	Interior
7	0.75	0.1875	Interior	19	−1.75	−1.6250	Interior
8	0.25	−2.3125	Interior	20	−1.00	2.1250	Interior
9	0.50	−1.0625	Interior	21	−0.25	4.6875	Top
10	0.00	−3.5625	Interior	22	0.00	4.7500	Top
11	−0.50	−4.8750	Bottom	23	1.00	5.0000	Top

Table 1: Displacement, force and state at each step.

A Python Code

```
import numpy as np
import matplotlib.pyplot as plt



# param_a: 0 = interior, 1 = top border, -1 = bottom border
# k2=ks

def bilinear(dy, fy, k2, x_ini, f_ini, dx, param_a):
    kp = fy / dy
    qy = fy - k2 * dy
    x_new = x_ini + dx
    f_top = qy + k2 * x_new
    f_bot = -qy + k2 * x_new

    if param_a == 0:
        df = kp * dx
        if f_ini + df > f_top:
            return f_top - f_ini, 1
        if f_ini + df < f_bot:
            return f_bot - f_ini, -1
```

```
        return df, 0

    if param_a == 1: # top
        if dx >= 0:
            return k2 * dx, 1
        df = kp * dx
        if f_ini + df < f_bot:
            return f_bot - f_ini, -1
        return df, 0

    if param_a == -1: # bottom
        if dx <= 0:
            return k2 * dx, -1
        df = kp * dx
        if f_ini + df > f_top:
            return f_top - f_ini, 1
        return df, 0

# TEST
dy, fy, k2 = 1.0, 5.0, 0.25
x = np.array([0, 0.5, 0.75, 1.25, 1.5, 1.0, 1.75, 0.75, 0.25, 0.5, 0.0, -0.5,
              -0.75, -1.5, -1.0, -1.75, -2.0, -2.5, -2.25, -1.75, -1.0, -0.25, 0, 1.0])

f = np.zeros(len(x))
param_a = 0

for i in range(1, len(x)):
    df, param_a = bilinear(dy, fy, k2, x[i-1], f[i-1], x[i]-x[i-1], param_a)
    f[i] = f[i-1] + df

dt = 1.0
t = np.arange(len(x)) * dt


plt.figure()
plt.axhline(0, color="black", lw=1)
plt.axvline(0, color="black", lw=1)
plt.plot(x,f)
plt.title("Part 1 - Hysteresis")
plt.xlabel("Displacement")
plt.ylabel("Force")
plt.show()

plt.figure()
plt.plot(t,f)
plt.title("Part 1 - Time-histories of displacement, Force")
plt.xlabel("Time")
plt.ylabel("Force")
plt.show()
```